

Lab Diagram — Device Functions

Reference companion to the tabloid diagram

Starlink-Primary Resilient Gateway Reference Lab • Digi International • July 2026

Companion to [assets/lab-diagram-tabloid-EN.pdf](#). It lists **what each device in the diagram does**. This document does **not** modify the diagram; it only explains it.

Convention: one site in the sprint (Ring 0 = Site A); three sites + central control in the expansion (Rings 1–2). All base hardware is identical between sites; only the profile (native / bonding / candidate) rotates between units.

Central Control — cloud + lab host (Proxmox) • AS65000

Device	Function
Digi Remote Manager Premier (cloud)	Single management plane: golden config + per-site settings, alerts (Device Offline / noncompliance), Push Monitor webhook → SIEM, Digi Containers deployment, and CCCS OTA for the embedded sensor.
Core / route collector	Public overlay endpoint (IKEv2/IPsec) and eBGP peer for every site (AS65000). Collects the network each site originates.
SIEM	Receives Push Monitor webhooks and the MP255 agent stream; event / security correlation.
Impairment platform	Injects controlled loss / jitter / latency to reproduce Starlink conditions during tests.
WAN Bonding server	External Tier 3 (1 Gb/s) bonding server required by the WAN-bonding experiment (A/B vs SureLink).
Lighthouse Enhance VM (Phase 2)	Opengear OOB management plane. Only needed if the OOB device is the Opengear OM1304 ; with a Digi Connect IT 4 the OOB is managed by Digi Remote Manager instead (no separate VM).

Site A — AS65001 • NATIVE profile • Ring 0 (2-week sprint)

Device	Function
Starlink Performance Kit (Gen 3)	Primary WAN. Copper to IX40 WAN/ETH1; delivers CGNAT 100.64.0.0/10 (public IPv4 optional).
Carrier A — 5G	In-band backup WAN through the IX40 5G modem; own SIM/APN.
Digi IX40-05	The resilient gateway. Runs SureLink failover, the IKEv2/IPsec overlay, eBGP (AS65001) and the Digi Container.
Digi Container	Metrics collector running on the IX40 under resource limits (proves Containers without degrading forwarding).
Service fabric (switch)	Service network behind SFP/ETH5 (1 Gb/s SFP, never SFP+): IPv4 /24 + IPv6 /64 advertised via eBGP.
Digi XBee Hive	RF gateway; isolates the sensor application from packet forwarding.
Digi XBee 3 (Zigbee)	Zigbee radio linking the Hive to the physical sensor.
Physical sensor	Measures temperature / door state / a digital input; not part of critical forwarding.

Device	Function
Digi ConnectCore MP255	Embedded SoM. DEY (Yocto) image; agent → SIEM (offline buffer + replay); OTA via CCCS.
OOB console server (<i>Phase 2</i>) — Opengear OM1304-4E-C-LSP and/or Digi Connect IT 4	Out-of-band rescue: serial console (DB9, Login mode) to the IX40, independent 5G Carrier B , management switch and independent power. Two interchangeable options — see the note below. Candidate to deploy at Site B .
Carrier B — 5G	Independent carrier feeding the OOB path (different operator and power from Carrier A / Starlink).
Primary PDU/UPS	Powers Starlink + IX40 + load.
Secondary PDU/UPS	Powers the OOB device + modem B (independent power domain).

Sites B & C (expansion • Rings 1–2)

Device	Function
Site B — AS65002	Identical hardware to Site A. BONDING profile: Starlink + 5G inside the bonded tunnel. Planned OOB via Digi Connect IT 4 .
Site C — AS65003	Identical to Site A. CANDIDATE / CHAOS profile: candidate containers, config and firmware; A/B/C profile rotation.

OOB console server — two interchangeable options

Both do the same job in the diagram (serial console to the IX40 + independent 5G + own power). They differ mainly in the management plane:

	Opengear OM1304-4E-C-LSP	Digi Connect IT 4
Serial to IX40	DB9 ↔ RJ45, adapter 319015 (Opengear X2 pinout)	RJ45 RS-232, Digi/Cisco-pinout cable
Cellular	5G Carrier B	LTE Cat 4 Carrier B (Digi CORE modem)
Mgmt ports	4-port mgmt switch	2× Ethernet 10/100 (add a small switch for 4 ports)
Managed by	Opengear Lighthouse (needs the Phase-2 VM)	Digi Remote Manager (same tenant as the routers → one plane)

Under consideration: Connect IT 4 at Site B. Choosing it unifies management under DRM and removes the Lighthouse VM; keeping the OM1304 preserves vendor-diverse OOB. Both are listed on purpose — the diagram is left unchanged for now.

Link legend (as drawn)

- Starlink primary WAN / overlay + eBGP (thick blue)
- 5G in-band backup / inter-site links
- Service fabric (1 Gb/s SFP fiber)
- Management / LAN — HTTPS, API, DRM (dotted)
- Out-of-band / Phase 2 elements (dashed)

-  Zigbee / cellular RF
- Dashed box = Phase 2

Source: `assets/lab-diagram-tabloid-EN.tex` and `03-arquitectura.md`. Product logos are property of their respective owners.